
SoundMatch-SR: The Rendering Transformation Is Learnable, the Taxonomy Is Not—A Dissociation in Synthetic–Real Sound-Effect Retrieval

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<https://github.com/elliottash/soundmatch-sr>

Abstract

Perceptual studies report that listeners cannot reliably distinguish a well-made Foley or text-to-audio rendering of an event from a real recording of the same event [Nordahl et al., 2010]: the *identity* of the event survives the rendering. I ask whether modern audio embeddings represent that identity invariantly to how a sound was produced, and I find a sharp *dissociation*. I introduce **SoundMatch-SR**, a benchmark for cross-domain (synthetic–real) sound-effect retrieval, built on (i) the category-aligned DCASE-2023 Task-7 Foley corpus [Choi et al., 2023] (5,550 real and 25,900 synthetic clips from 37 generator systems, 7 classes) and (ii) a new 34-category benchmark labelled with the Universal Category System (FSD50K [Fonseca et al., 2022] real audio, CLAP-verified, with 10,420 instance-level Stable-Audio-Open [Evans et al., 2024b] twins). Off-the-shelf encoders carry a real, linearly-accessible synthetic–real gap (CLAP: 0.161 mAP; Proxy-A-distance 1.27). A lightweight head on frozen embeddings closes it *within* the training taxonomy (0.161 $\rightarrow$ 0.018), but a *class-supervised* head does *not* transfer to unseen categories: held-out classes collapse below the frozen baseline, on both 7 and 34 categories. The remedy is to train on the *pairs*, not the labels: an instance-contrastive objective (a clip and its generated twin) learns the rendering *mapping* rather than category identity, and generalizes—on categories never seen in training it retrieves the exact real twin with $R@1 = 0.83 \pm 0.05$ (vs. 0.64 frozen, 0.33 class-supervised), across five folds and three encoders (CLAP, PANNs, AST). Category *clustering* does not transfer for any objective. I release the benchmark, the corpus recipe, the trained heads, and a reusable embedding transform.

1 Introduction

Foley research finds that well-synthesized everyday sounds are perceptually indistinguishable from recordings of the same action; the event’s identity survives the synthesis-to-recording boundary for a human listener [Nordahl et al., 2010]. I make this measurable for machine representations:

Is “sound identity” recoverable in embedding space, invariant to the rendering process—and if so, does that invariance transfer to sounds the model was not trained on?

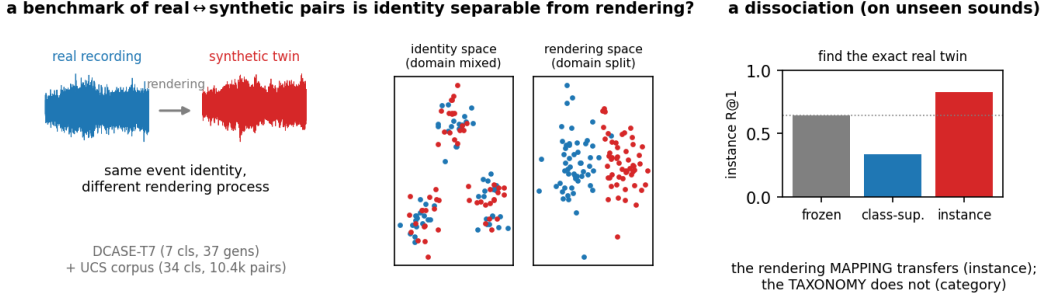


Figure 1: **SoundMatch-SR**. I pair each real recording with a synthetic twin of the same event and ask whether an embedding can represent the event *identity* invariantly to the rendering. The synthetic-to-real *mapping* transfers to sounds never seen in training (instance retrieval, right), while category identity does not.

I operationalize it as retrieval rather than detection: detection rewards *exploiting* the synthetic–real gap, whereas retrieval and invariance reward *removing* it, which is the useful capability for sound-effect search, real/synthetic deduplication, and generative-audio evaluation. My answer is a dissociation that, to my knowledge, is new: the synthetic-to-real *transformation* is a learnable, transferable object, while category identity is not.

Contributions. (C1) a two-part benchmark—a controlled 7-class corpus (DCASE-T7) and a diverse, instance-paired 34-category corpus—with leakage-safe splits and a reproducible generation recipe; (C2) a measured dissociation: an instance-contrastive objective generalizes the synthetic-to-real mapping to unseen categories, whereas class-supervised invariance does not (and can degrade unseen classes below baseline), robust over five folds and three encoders; (C3) a fidelity-spectrum analysis showing the instance head recovers the mapping across twin-fidelity levels, collapsing only when the twin is independent of its source; (C4) released artifacts—benchmark, recipe, CLAP-verification, trained heads, and a reusable transform that adjusts any new clip’s embedding.

2 Related Work

Audio-representation benchmarks. HEAR [Turian et al., 2022] and the recent massive embedding benchmarks MSEB [Heigold et al., 2026] and MAEB [El Assadi et al., 2026] evaluate frozen embeddings across classification, retrieval, and clustering, but treat all audio as one domain; none tests invariance to *how* a sound was produced. The encoders I benchmark—CLAP (Wu et al., 2023; Elizalde et al., 2023), PANNs [Kong et al., 2020], AST [Gong et al., 2021], and the SSL families BEATs [Chen et al., 2023], M2D [Niizumi et al., 2024], AudioMAE [Huang et al., 2022]—are evaluated frozen.

Synthetic-audio detection. Deepfake-environmental-audio detection is built on exactly the DCASE-T7 corpus [Ouaïdi et al., 2024] and benchmarked in EnvSDD [Yin et al., 2025]; this pursues the mirror objective (exploit the gap). I remove it, and reproduce detection as a controlled *sensitive* head.

Text-to-audio generators. The synthetic domain comes from modern generators—Stable Audio Open [Evans et al., 2024b] and its predecessor [Evans et al., 2024a], AudioGen [Kreuk et al., 2023], AudioLDM [Liu et al., 2023]—whose realism is typically scored by Fréchet Audio Distance [Kilgour et al., 2019]; my sensitive head offers a complementary, paired realness axis.

Domain adaptation and contrastive learning. I draw on Proxy-A-distance [Ben-David et al., 2010], domain-adversarial training [Ganin and Lempitsky, 2015], Deep CORAL [Sun and Saenko, 2016], supervised contrastive learning [Khosla et al., 2020], and IRM [Arjovsky et al., 2019]. My generalizing objective is instance-level in the sense of instance discrimination [Wu et al., 2018] and contrastive representation learning (Chen et al., 2020b; Radford et al., 2021); the negative result—that class-level supervision does not transfer the invariance—is a domain-generalization phenomenon.

Table 1: The two heads on DCASE-T7 (left) and the bridging ablation (right). The invariant head closes the gap and organises the space by event; the sensitive head is its mirror. Cross-domain supervised contrast does the work.

head	control	cross mAP	sil(ev/dom)	ablation	gap	PAD
frozen	0.935	0.774	0.18 / 0.04	supcon only	+0.019	1.38
invariant	0.988	0.970	0.75 / 0.00	+CORAL	+0.017	1.38
sensitive	0.206	0.157	−0.04 / 0.75	+DANN	+0.019	1.38
				+DANN+IRM	+0.018	1.38

3 The SoundMatch-SR Benchmark

Core corpus (DCASE-T7). The DCASE-2023 Task-7 corpus [Choi et al., 2023] is a ready-made, category-aligned real/synthetic set across 7 classes: 5,550 real clips (sourced from FSD50K, UrbanSound8K [Salamon et al., 2014], BBC, and Freesound; I use DCASE’s source-disjoint dev/eval as train-val/test) and 25,900 synthetic clips from 37 generator systems. It is the controlled setting for a clean gap measurement and a leave-one-class-out probe.

UCS corpus. To test generalization I need many categories and instance-level pairs. I label real audio with the Universal Category System (UCS), mapping the 200 AudioSet [Gemmeke et al., 2017] labels of FSD50K onto 34 UCS categories spanning all morphologies (impact, texture, tonal, whoosh, vocal, mechanical, ambience). I *verify* every clip with zero-shot CLAP [Wu et al., 2023] (keep iff its label is top-5 by audio–text cosine), retaining 10,420/13,579 (77%) and removing FSD50K’s noisiest multi-labels. For each verified anchor I generate a Stable-Audio-Open *audio-init* twin [Evans et al., 2024b] conditioned on the real clip, yielding 10,420 instance pairs. Splits follow FSD50K’s train/val/eval, keyed on the Freesound uploader so crops of one recording never straddle the boundary.

Tasks and metrics. *Category retrieval*: query a clip; rank opposite-domain clips; relevant iff same category. I report mAP and a same-domain control, so the headline *domain gap* = control − cross. *Instance retrieval* (UCS): relevant iff the exact cross-domain twin (shared instance id); R@1 and MRR. I report bootstrap CIs, and on the UCS corpus a *k*-fold leave-classes-out protocol (Sec. 5). Full per-class, per-split statistics and the datasheet are in the appendix.

4 Method

I train a small MLP head (CLAP-512 $\rightarrow$ 256, ℓ_2 -normalised) on *frozen* embeddings; the head is the adjusted embedding and applies to any new clip. The objective decides what it learns. The **invariant** head is class-supervised contrastive with cross-domain positives up-weighted (optionally with DANN/IRM): it pulls same-category clips together and removes domain. The **sensitive** head is domain-supervised within category: the deliberate mirror, organising the space by rendering—its real-vs-synthetic direction is a fidelity axis. The **instance** head uses as positive a clip and *its own* generated twin (with pair-aware batching so twins co-occur in a batch); it learns the synthetic-to-real *mapping* rather than class identity.

5 Experiments

Frozen encoders carry a real gap (Table 2). The synthetic–real axis is real and linearly accessible in both CLAP and PANNs [Kong et al., 2020].

Within-taxonomy the gap closes; the two heads are mirror images. The invariant head closes the DCASE gap 0.161 $\rightarrow$ 0.018 while *raising* same-domain control (0.935 $\rightarrow$ 0.988) and making the space event-organised (silhouette by event 0.18 $\rightarrow$ 0.75, by domain 0.04 $\rightarrow$ 0.00); the sensitive head flips it to organise by rendering (PAD 1.88, silhouette by domain 0.75). Cross-domain supervised contrast alone closes the gap; DANN/CORAL/IRM add < 0.002 (Table 1); the geometry flip is visible in Fig. 2.

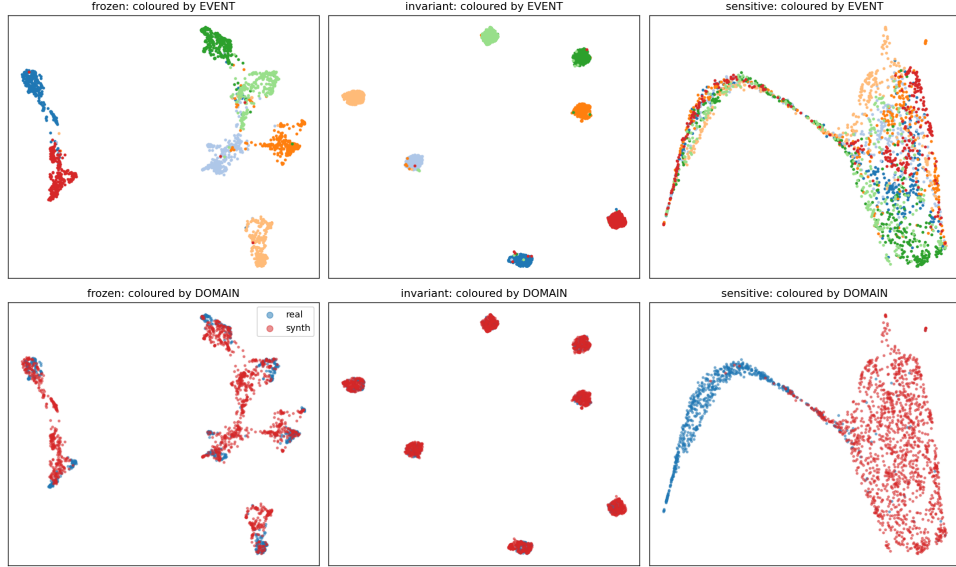


Figure 2: UMAP of the DCASE-T7 test set, frozen vs. the two heads, coloured by event (top) and by domain (bottom). The invariant head tightens event clusters with real/synth overlapping within each; the sensitive head splits real from synth and smears events—the geometry flips.

Table 2: Frozen encoders: synthetic–real gap on DCASE-T7 (test split).

frozen	control	synth→real mAP	gap	PAD	domain-probe
CLAP	0.935	0.774 [0.766,0.783]	+0.161	1.27	0.90
PANNs CNN14	0.798	0.674	+0.124	1.39	0.90

The closed-world failure. Holding one class out of training and evaluating on it, the held-out class collapses *below* frozen (keyboard $0.69 \rightarrow 0.43$, gunshot $0.81 \rightarrow 0.31$, rain $0.75 \rightarrow 0.30$). Adding categories does not fix it (34-class: $0.40 \rightarrow 0.19$).

The dissociation (Table 3, Fig. 3a). Training on the pairs, the instance objective generalizes the mapping to unseen categories (instance-R@1 0.83, every fold), class-supervision *hurts* it (0.33), and no objective transfers category clustering (all $\leq$ frozen on unseen-category mAP). It holds on three encoders—CLAP, PANNs, and AST [Gong et al., 2021] (instance R@1 0.83/0.75/0.95 vs. class 0.33/0.38/0.38)—so it is a property of the objective, not of any one encoder.

Not near-copies (Fig. 3b). Across twin-fidelity levels (measured as the CLAP cosine of a twin to its source) the instance head holds up (advantage $+0.43..0.52$ over frozen) and collapses to chance only when the twin is independent of its source—so it recovers a rendering mapping, not a near-copy shortcut.

Cross-generator scope. Adding 670 ElevenLabs (text-only) twins captioned from each clip’s metadata, instance correspondence is weak (frozen R@1 0.11 vs. 0.55–0.98 for audio-init), because the caption is a bottleneck: the twins are category-appropriate but not renderings of the *specific* clip. Instance-level pairing thus requires the generator to condition on the source audio.

6 Conclusions, Limitations, and Future Work

The synthetic-to-real rendering transformation is a learnable, transferable object; category identity is not. **Limitations:** category clustering does not transfer to unseen categories (train per target taxonomy); twins are Stable-Audio audio-init at one operating point per corpus; frozen-embedding heads cannot recover information the backbone discarded; instance pairing requires audio-conditioned

Table 3: Five-fold leave-classes-out on the UCS corpus: held-out (unseen) categories, mean $\pm$ std.

variant (unseen categories)	category-mAP	instance-R@1	instance-MRR
frozen	0.615 $\pm$ 0.052	0.642 $\pm$ 0.094	0.731 $\pm$ 0.081
class-supcon	0.459 $\pm$ 0.072	0.334 $\pm$ 0.064	0.439 $\pm$ 0.061
instance	0.492 $\pm$ 0.047	0.829 $\pm$ 0.045	0.888 $\pm$ 0.034

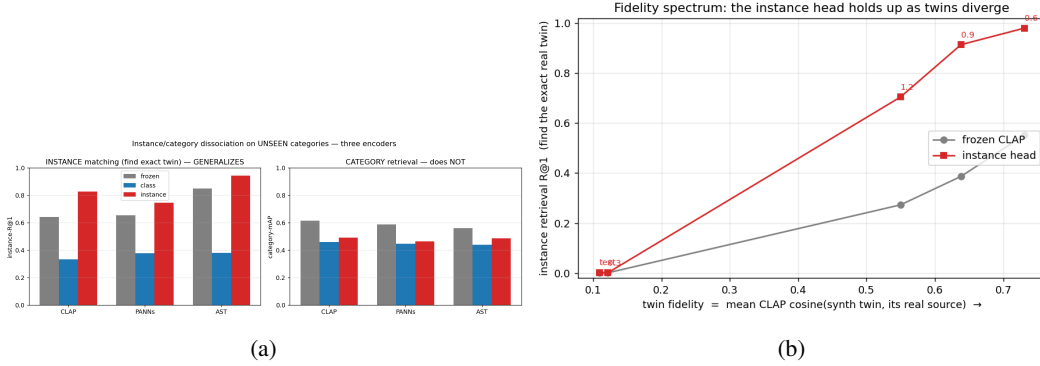


Figure 3: (a) The instance/category dissociation on unseen categories, three encoders: instance matching generalizes (left), category retrieval does not (right). (b) Fidelity spectrum: the instance head holds up as twins diverge from their source.

generation. SSL encoders (BEATs [Chen et al., 2023], AudioMAE [Huang et al., 2022]) are provided as activatable loaders. **Future work:** broader real sources (e.g. VGGSound [Chen et al., 2020a]) and domain-relevant corpora (game and film SFX), and using the sensitive head as a realness metric for generative-audio evaluation.

Reproducibility and release

Code (MIT), DCASE and UCS manifests, the CLAP-verification, the generation recipe (event, prompt, seed, steps—bit-reproducible without redistributing restricted audio), the trained heads, and a reusable transform are released. A datasheet [Gebu et al., 2021] and a persistent hosting/maintenance plan are in the appendix.

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A Datasheet, hosting, and maintenance

The appendix provides the Gebru et al. [Gebru et al., 2021] datasheet (motivation, composition, collection, preprocessing, uses, distribution), per-class/per-split statistics, the licensing matrix (CC-licensed audio redistributed; BBC and restricted audio shipped as a fetch manifest plus the generation recipe), and the maintenance plan (versioned manifests; the code regenerates all derived artifacts from the manifest and the source archives).

B Corpus statistics and per-event analysis

Table 4: Per-domain, per-split clip counts. DCASE-T7 real test is DCASE’s source-disjoint eval set; the UCS corpus uses FSD50K’s train/val/eval, uploader-keyed.

corpus	domain	train	val	test
DCASE-T7	real	4,150	700	700
DCASE-T7	synth	18,165	3,884	3,851
UCS (34 cls)	real	8,315	999	4,265
UCS (34 cls)	synth (twins)	6,636	719	3,065

Table 5: Per-event frozen-CLAP cross-domain (synth→real) mAP on DCASE-T7. Fast, granular transient/mechanical sounds transfer worst.

	motor	gunshot	dog_bark	sneeze	rain	footstep	keyboard
mAP	0.852	0.810	0.809	0.793	0.752	0.718	0.692

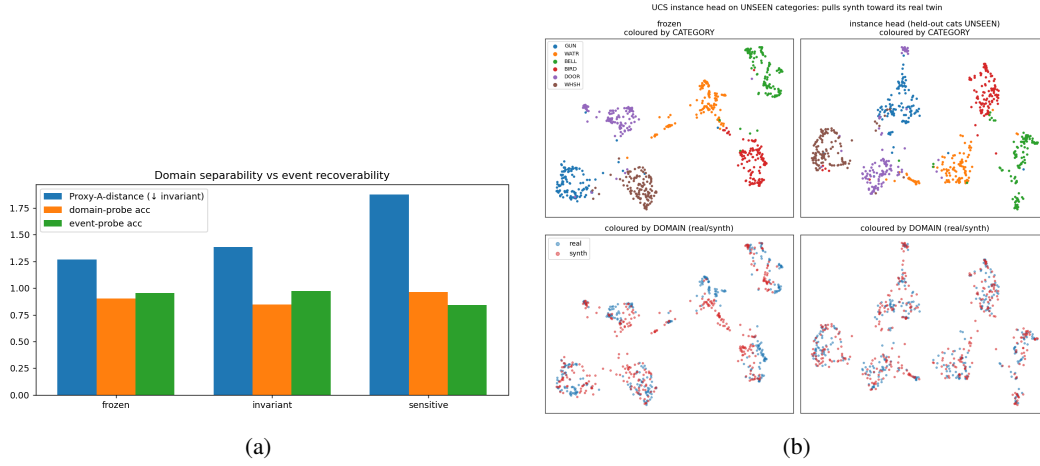


Figure 4: (a) Proxy-A-distance and event/domain linear-probe accuracy across the DCASE heads. (b) UMAP of the instance head on unseen UCS categories: it mixes real/synth (domain) more than frozen.